HW4 B1121123 巫煊崴

程式碼

```
‘timescale 1ns/10ps
‘define WL 9
‘include "Hw4_3bitCLA_rtl - Student.v"

module Hw4_9bitCLA (
    input [‘WL-1:0] X,
    input [‘WL-1:0] Y,
    input Cin,
    output [‘WL-1:0] Sum,
    output Cout,
    output OV
);

    wire [2:0] G_LCG;
    wire [2:0] P_LCG;
    wire [3:0] Carry_in;

    assign Carry_in[0] = Cin;
    assign Carry_in[1] = G_LCG[0] |  (P_LCG[0] & Carry_in[0]);
    assign Carry_in[2] = G_LCG[1] |  (P_LCG[1] & G_LCG[0]) |  (P_LCG[1] & P_LCG[0] & Carry_in[0]);
```

```
    assign Carry_in[3] = G_LCG[2] |  (P_LCG[2] & G_LCG[1]) |  (P_LCG[2] & P_LCG[1] & G_LCG[0]) |  (P_LCG[2] &
P_LCG[1] & P_LCG[0] & Carry_in[0]);
    assign Cout = Carry_in[3];

    Hw4_3bitCLA CLA_1 (
        .X(X[2:0]),
        .Y(Y[2:0]),
        .Cin(Carry_in[0]),
        .Sum(Sum[2:0]),
        .G(G_LCG[0]),
        .P(P_LCG[0])
    );

    Hw4_3bitCLA CLA_2 (
        .X(X[5:3]),
        .Y(Y[5:3]),
        .Cin(Carry_in[1]),
        .Sum(Sum[5:3]),
        .G(G_LCG[1]),
        .P(P_LCG[1])
    );

    Hw4_3bitCLA CLA_3 (
```

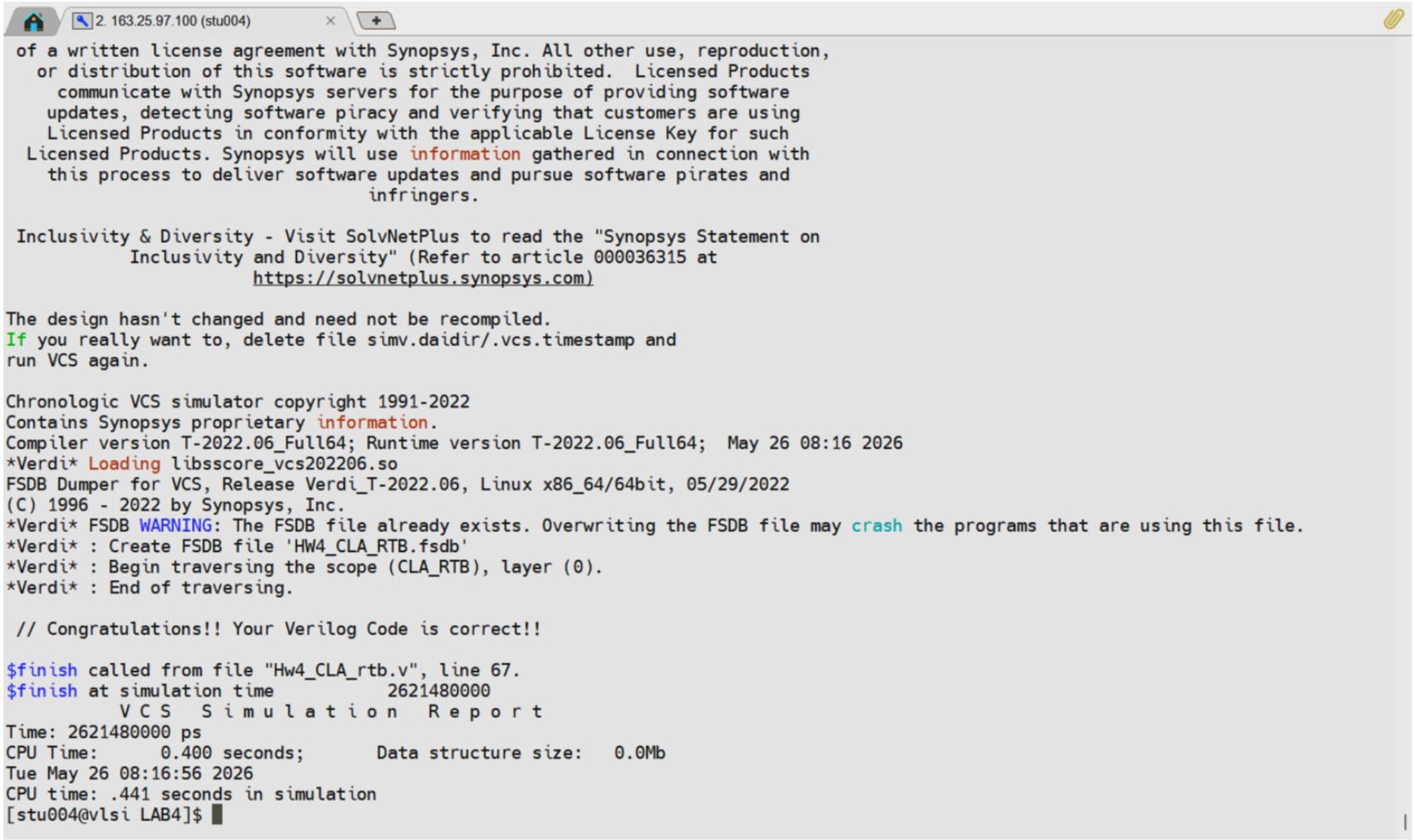
```
        .X(X[8:6]),
        .Y(Y[8:6]),
        .Cin(Carry_in[2]),
        .Sum(Sum[8:6]),
        .G(G_LCG[2]),
        .P(P_LCG[2])
    );

    // OV = (X_sign == Y_sign) && (Sum_sign != X_sign)
    assign OV = (X[`WL-1] ~^ Y[`WL-1]) & (Sum[`WL-1] ^ X[`WL-1]);

endmodule


`timescale 1ns/10ps
`include "Hw4_1bitCLA_rtl - Student.v"
module Hw4_3bitCLA (
    input [2:0] X,
    input [2:0] Y,
    input Cin,
    output [2:0] Sum,
    output  G, // Group Generate
    output  P  // Group Propagate
```

```
);

    wire [2:0] G_LCG;   // 3-bit lookahead carry generate from sub-modules
    wire [2:0] P_LCG;   // 3-bit lookahead carry propagate from sub-modules
    wire [2:0] Carry_in;


    assign Carry_in[0] = G_LCG[0] | (P_LCG[0] & Cin);
    assign Carry_in[1] = G_LCG[1] | (P_LCG[1] & G_LCG[0]) | (P_LCG[1] & P_LCG[0] & Cin);
    assign Carry_in[2] = G_LCG[2] | (P_LCG[2] & G_LCG[1]) | (P_LCG[2] & P_LCG[1] & G_LCG[0]) | (P_LCG[2] & P_LCG[1]
& P_LCG[0] & Cin);


    Hw4_1bitCLA cla_1 (
        .X(X[0]),
        .Y(Y[0]),
        .Cin(Cin),
        .Sum(Sum[0]),
        .G(G_LCG[0]),
        .P(P_LCG[0])
    );

    Hw4_1bitCLA cla_2 (
```

```
        .X(X[1]),
        .Y(Y[1]),
        .Cin(Carry_in[0]),
        .Sum(Sum[1]),
        .G(G_LCG[1]),
        .P(P_LCG[1])
    );

    Hw4_1bitCLA cla_3 (
        .X(X[2]),
        .Y(Y[2]),
        .Cin(Carry_in[1]),
        .Sum(Sum[2]),
        .G(G_LCG[2]),
        .P(P_LCG[2])
    );

    assign P = P_LCG[2] & P_LCG[1] & P_LCG[0];
    assign G = G_LCG[2] | (P_LCG[2] & G_LCG[1]) | (P_LCG[2] & P_LCG[1] & G_LCG[0]);

endmodule
```

```
‘timescale 1ns/10ps
module Hw4_1bitCLA (
    input X,
    input Y,
    input Cin,
    output Sum,
    output G,
    output P
);

assign P = X ^ Y;
assign G = X & Y;
assign Sum = P ^ Cin; //  X ^ Y ^ Cin

endmodule
```

模擬結果


2. 163.25.97.100 (stu004)
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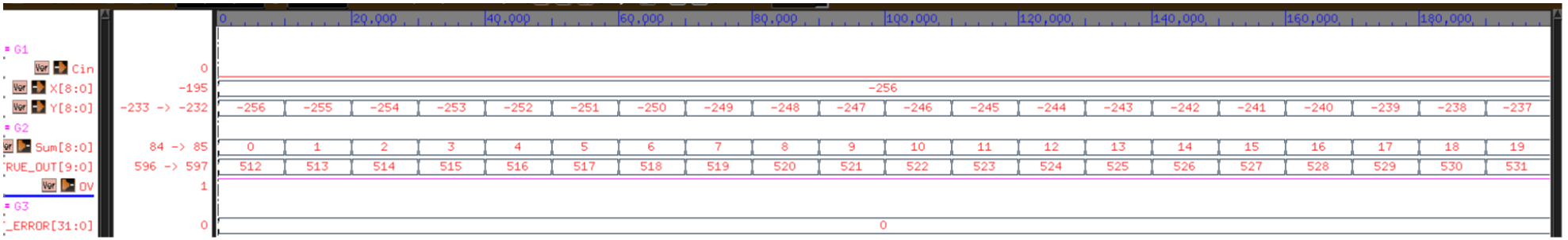
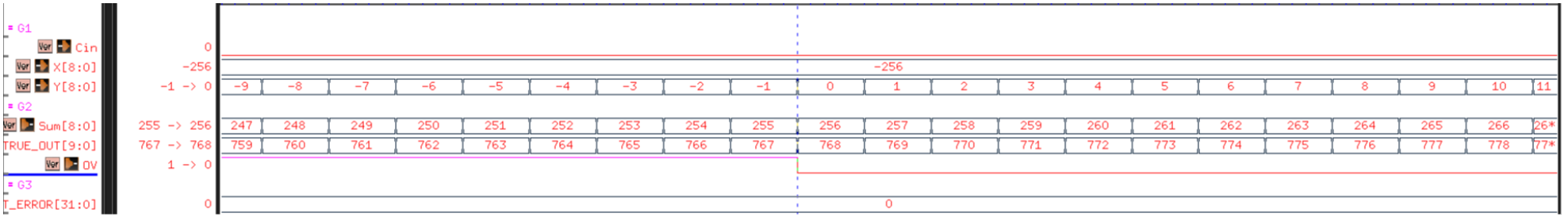
Inclusivity & Diversity - Visit SolvNetPlus to read the "Synopsys Statement on
Inclusivity and Diversity" (Refer to article 000036315 at
https://solvnetplus.synopsys.com)
The design hasn't changed and need not be recompiled.
If you really want to, delete file simv.daidir/.vcs.timestamp and
run VCS again.
Chronologic VCS simulator copyright 1991-2022
Contains Synopsys proprietary information.
Compiler version T-2022.06_Full64; Runtime version T-2022.06_Full64; May 26 08:16 2026
Verdi Loading libsscore_vcs202206.so
FSDB Dumper for VCS, Release Verdi_T-2022.06, Linux x86_64/64bit, 05/29/2022
(C) 1996 - 2022 by Synopsys, Inc.
Verdi FSDB WARNING: The FSDB file already exists. Overwriting the FSDB file may crash the programs that are using this file.
Verdi : Create FSDB file 'HW4_CLA_RTB.fsdb'
Verdi : Begin traversing the scope (CLA_RTB), layer (0).
Verdi : End of traversing.
// Congratulations!! Your Verilog Code is correct!!
$finish called from file "Hw4_CLA_rtb.v", line 67.
$finish at simulation time 2621480000
V C S Simulation Report
Time: 2621480000 ps
CPU Time: 0.400 seconds; Data structure size: 0.0Mb
Tue May 26 08:16:56 2026
CPU time: .441 seconds in simulation
[stu004@vlsi LAB4]$

波形圖


G1
Cin
X[8:0]
Y[8:0]
G2
Sum[8:0]
TRUE_OUT[9:0]
OV
G3
T_ERROR[31:0]
0
-195
-233 -> -232
84 -> 85
596 -> 597
1
0
0
20,000
40,000
60,000
80,000
100,000
120,000
140,000
160,000
180,000
-256
-256 -255 -254 -253 -252 -251 -250 -249 -248 -247 -246 -245 -244 -243 -242 -241 -240 -239 -238 -237
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19
512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531
0


G1
Cin
X[8:0]
Y[8:0]
G2
Sum[8:0]
TRUE_OUT[9:0]
OV
G3
T_ERROR[31:0]
0
-256
-1 -> 0
255 -> 256
767 -> 768
1 -> 0
0
-256
-9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10 11
247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 26*
759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 77*
0